J Behav Educ (2015) 24:459–469
DOI 10.1007/s10864-015-9224-1

COMMENTARY

What Works Clearinghouse Standards and Generalization of Single-Case Design Evidence

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Published online: 14 March 2015

Abstract A recent review of existing rubrics designed to help researchers evaluate the internal and external validity of single-case design (SCD) studies found that the various options yield consistent results when examining causal arguments. The authors of the review, however, noted considerable differences across the rubrics when addressing the generalization of findings. One critical finding is that the What Works Clearinghouse (WWC) review process does not capture details needed for report readers to evaluate generalization. This conclusion is reasonable if considering only the WWC's SCD design standards. It is important to note that these standards are not used in isolation, and thus generalization details cannot be fully understood without also considering the review protocols and a tool called the WWC SCD review guide. Our purpose in this commentary is to clarify how the WWC review procedures gather information on generalization criteria and to describe a threshold for judging how much evidence is available. It is important to

Some of the information contained herein is based on the What Works Clearinghouse's *Single-case design technical documentation version 1.0* (Pilot) (referred to as the *Standards* in this article) produced by two of the current authors (Kratochwill and Hitchcock) and the Panel members and available at http://ies.ed.gov/ncee/wwc/pdf/wwc_scd.pdf. The *Standards* that are described in the technical documentation were developed by a Panel of authors for the Institute of Education Sciences (IES) under Contract ED-07-CO-0062 with Mathematica Policy Research, Inc. to operate the What Works Clearinghouse (WWC). The content of this article does not necessarily represent the views of the Institute of Education Sciences or the WWC.

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clarify how the system works so that the SCD research community understands the standards, which in turn might facilitate use of future WWC reports and possibly influence both the conduct and the reporting of SCD studies.

Keywords Single-case design · Generalization · Internal validity · External validity

There is a long-standing call for using interventions with a strong evidence base (e.g., Deegear and Lawson 2003; Kratochwill 2002; Kratochwill and Stoiber 2000; Schneider et al. 2007). One set of methodologies that has been recognized as a viable approach for generating empirical evidence to inform treatment innovation, adoption, or improvement is the single-case design (SCD) approach (e.g., Horner et al. 2005; Kazdin 2011; Kratochwill and Levin 2014). SCDs are experimental methods consisting of various designs involving repeated measures of a specific behavior or skill under different conditions to evaluate the effectiveness of a treatment for an individual or a small group of individuals that serve as their own control (Kazdin 2011). SCDs have emerged from the field of psychology and have been used across various disciplines including education, medicine, and speech and language therapy.

Like most investigations, one SCD study is unlikely to generate sufficient empirical evidence to warrant policy change, even if it might compel alteration to localized practice. Thus, it is important to not only evaluate the results of a single SCD but also synthesize evidence from multiple SCD studies examining the effectiveness of a treatment and then make inferences about generalizing findings to a population of interest, as well as potentially to other populations and settings. Collating evidence from multiple SCD studies conducted by different research teams, with different participants, and across different settings has the potential to provide stronger evidence that might inform treatment decisions and policy change.

A key part of any effort to collate empirical evidence is to generate rubrics that can be used to judge the findings of individual studies and subsequently summarize information in the form of systematic reviews. Recently, researchers have published a number of rubrics, or guidelines, for judging SCD evidence (e.g., Kratochwill et al. 2010, 2013; Smith 2012; Wendt and Miller 2012) and conducted SCD systematic reviews (e.g., Bowman-Perrott et al. 2013; Dart et al. 2014). Maggin et al. (2013) have made an important contribution to such efforts by comparing seven different rubrics designed to assist scientists, practitioners, or legislators in evaluating findings of SCD studies. Maggin et al. examined the consistency of existing rubrics designed to assess different requirements of SCD methodology related to internal and external validity. They first reviewed each rubric and then applied each one to a set of SCD studies focusing on self-management interventions.

Their effort yielded a number of key findings, one of which was the consistency of internal validity judgments made about component SCD studies across the seven rubrics.[1] A second key finding was that there was limited agreement across the rubrics pertaining to issues of generalizing evidence. According to the Maggin et al., some rubrics were designed to capture considerable detail about generalization, whereas other rubrics functionally ignored this consideration. For example, Maggin et al. state: "…the WWC criteria provided guidance solely for criteria related to establishing experimental control while others included several descriptive criteria related to establishing the generality of the intervention" (p. 20).

This particular conclusion about the WWC represents the motivation for this commentary. The WWC criteria and review procedures do in fact deal with documenting information that informs generalization and there is value in explaining to stakeholders how these procedures are applied when conducting a review. But at the outset, it is important to note that we understand that several factors may contribute to the confusion about the WWC *Pilot Standards* (i.e., the *Standards*) and, in particular, how generalization of findings is handled. One contributing factor to the confusion about the *Standards* and generalization relates to the fact that Maggin et al. (2013) may have reviewed the *Standards* document in isolation of review protocols and the SCD *Review Guide*. The WWC review protocols are, however, important because they specify the research questions to be addressed via a review; they also describe the population(s) of interest, the relevant outcome domains, and the settings in which interventions should be applied. The above-mentioned aspects are critical because part of understanding generalization is thinking through the populations, settings, and contexts to which one might want to generalize information. The SCD *Review Guide* represents a database where all the relevant aspects of a study are first documented in a systematic manner and then evaluated to draw a conclusion about the evidence presented in a study. Another contributing factor to the confusion about *Standards* and generalization relates to the fact that WWC reporting on SCD evidence has been, to date, minimal. Therefore, limited information on the application of *Standards* to identifying empirical evidence across multiple SCD studies is publicly available, which perhaps lead some to assume that the *Standards* do not address this important issue.

Thus, our purpose in this commentary is to clarify the WWC procedures with particular focus on how the *Standards* are implemented to address the generalization of findings from SCD studies reviewed within the context of a specific protocol. We believe such clarification is worthwhile because an understanding of these procedures among the SCD research community will facilitate the use of future WWC project reports and may influence both the conduct and the reporting of studies that use these types of designs. We begin first with a description of the external and internal validity of SCDs. We then present different approaches to

[1] Shadish et al. (2002) argue that internal validity, or the degree to which a causal relationship exists between a treatment and outcome variable is valid, is the *sin qua non* of experimental design. In other words, there might not be much point in carefully pondering the external validity (which is related to generalization) of studies that do not yield strong evidence of a causal effect. This position is because, if one cannot demonstrate that a given treatment was responsible for some outcome, then there is little point in examining whether the evidence generalizes to different contexts. As applied to SCDs, if one has no or limited confidence that there is a functional relationship between a treatment (independent variable) and dependent variable, then why do the hard work of generalizing?

evaluate the generalization of findings from SCD studies (detailed discussion around evaluating experimental control is available in Kratochwill et al. 2010, 2013). Next, we discuss the *Standards* within the context of a WWC review protocol focusing on criteria used to address generalization including the *5-3-20* rule. We end by highlighting the importance of ongoing refinement of the *Standards* to better capture methodological criteria of SCD with the ultimate goal of informing policy and practice.

Internal and External Validity

As noted above, it appears from the Maggin et al. (2013) comparison that there was reasonable consistency across the rubrics pertaining to judgments of internal validity. This finding is not surprising. Through work in *Standards* development in the WWC and other ventures, such as the *Task Force for Evidenced-Based Interventions in School Psychology* (see Kratochwill and Stoiber 2002), it became clear that the Campbellian validity framework (Shadish 1995; Shadish et al. 2002) applies to a broad number of designs that are capable of yielding causal evidence. This evidence is generated from SCDs when they are used to evaluate treatment effects. The task of judging internal validity first entails specifying the causal questions at hand and then selecting the design that allows one to control for a common number of threats to internal validity, which in essence, represent alternative explanations for any observed changes to a dependent variable after treatment exposure. Examples of such threats are maturation, history, regression to the mean, diffusion of treatment, and instrumentation (see Shadish et al. 2002 for details). SCDs can be designed in such a way to render these alternative explanations as implausible. Identifying the presence of these design features will yield judgments about whether there is strong evidence that a treatment worked as intended (cf. Horner et al. 2005; Kratochwill et al. 2010, 2013). Thus, given the logic behind causal inference, we might expect that the sundry rubrics yield fairly consistent conclusions pertaining to internal validity.

In our experience, assessing external validity is a more complex prospect than judging internal validity. External validity refers to the extent to which causal inference from a particular study holds over different contexts, settings, measures, populations, and so on (Shadish et al. 2002) and may also be thought of as a broad facet of generalization. Similar to internal validity, a number of threats may limit the generalization of the findings of a SCD study. Examples of such threats include multiple-treatment interference (i.e., if an observed outcome was due to multiple and interacting treatments then the effect will not generalize), generality across settings, generality across subjects, and generality across outcomes (Kazdin 2011; Shadish et al. 2002). Threats of these sort deal with the basic question whether an observed effect from a study will hold over changes to subject characteristics, specifics in a setting, and similar but different types of outcomes. The challenge in evaluating the external validity of a given study is partially due to the fact that many factors or characteristics of an experiment may represent a threat to one's capacity to generalize, with some factors being easily identifiable, whereas others are not.

Another challenge when evaluating the generalization of findings is that a researcher conducting a review effort to collate evidence across multiple SCDs may not know the point to which consumers of information might wish to generalize.

A basic solution to address the above-mentioned situation is to first specify the research questions of interest in a review protocol (e.g., What treatments are effective at improving behavior among K-12 students classified with an emotional-behavioral disorder?). Articulating the key research questions of a review helps to frame generalization goals of the effort. Next, reviewers must consider different approaches to determining what type of information to include when collating evidence. One option is to include findings from all studies on a specific topic that were located, whether they are characterized by strong validity or not, and assess the totality of evidence pertaining to a logically grouped set of treatments.

Alternatively, researchers can use what Maggin et al. (2013) describe as a gating procedure. The first step of a gating procedure consists of identifying studies on a specific topic that will be included in the review. From there, only studies with strong internal validity are considered (i.e., studies must pass an internal validity *gate* before they are considered further). This process is applied when using the WWC *Standards* to code SCD studies; that is, studies are coded for evidence criteria only after they pass the design standards. As with most choices, there are trade-offs here. The former option entails reporting all evidence but may also yield some confusion for consumers because synthesis findings might have to explain that some evidence is not very strong or, in the case of the WWC *Standards*, do not meet design standards. This approach may become especially problematic when there is no clear overall picture (e.g., several studies with varying internal validity support treatment adoption and several do not). The gating procedure, by contrast, can yield findings that are easier to communicate because, whatever the results of component studies, they would all be characterized by reasonably strong internal validity. However, information is functionally barred from informing the review questions on the basis that, if a study is determined to have weak internal validity, it will not be included in the review. The WWC uses the gating procedure for SCDs as well as group-design studies (i.e., randomized controlled trials and quasi-experiments; see WWC 2013). This gating approach is arguably a reasonable one when the goal is to inform practitioners and policymakers, which is the expressed intent of the WWC.

Regardless of whether a review uses a gating procedure or not, the next step entails evaluating the external validity of the studies included in the review. Two complexities that come with considering SCD evidence is that questions of internal and external validity concerns are not always mutually exclusive. For example, one must see a detailed description of the baseline and treatment procedures to understand the contrast; yet, baseline details describe the status quo and thus inform generalization. As we mentioned previously, when evaluating the external validity of findings within the context of a gating procedure, the WWC faces the challenge of not knowing exactly the scenarios to which the report consumers might wish to generalize.

Nevertheless, the WWC addresses the external validity of findings from multiple SCD studies by taking into consideration what Maggin et al. (2013) describe as criteria for assessing generality. Maggin et al. argue that “Single-case research…requires the collection and careful reporting of critical aspects of the research including information pertaining to participant characteristics, setting procedures, baseline conditions, and operational definitions of the variables being studied" (p. 6). One reason for providing a detailed description of these aspects is to allow readers to understand situations to which findings might be generalized. For example, the *Participant Information* provides information about the demographic and individual characteristics of people included in a study and allows consumers to make inferences on the extent to which people not included in the study may benefit from a treatment.

The *Setting Description* provides details about the context in which the research was conducted while allowing readers to evaluate the extent to which the treatment may be effective when applied in a different context than the one in which the study was conducted. It is also important to consider the description of baseline procedures (i.e., the *Baseline Description* criterion) so readers understand the treatment contrast being examined by a SCD study and support generalization and replication of findings. The two key variable types, *Independent* and *Dependent,* must also be described in detail. The former essentially describes the treatment or intervention examined and the latter deals with outcome variables of interest (e.g., behavior, skill).

We concur that examining and reporting of the above-mentioned criteria is necessary and the WWC review process captures these aspects when evaluating SCD studies. The ultimate goal of this process is describing the treatment with sufficient detail so that practitioners and policymakers can make their own decisions about whether the available evidence applies to their circumstances. Moreover, the WWC procedure also takes things a step further by applying a novel threshold for determining whether SCD evidence has been sufficiently repeated or replicated to warrant generation of a report. This threshold is the 5-3-20 rule. In the next section, we describe the review procedures and the threshold so that readers have a better understanding of how the WWC assesses generality of SCD evidence, thus clarifying any potential misunderstanding related to generalization of findings within the context of *Standards.* We also hope that this provides clear guidelines for researchers who may be interested in conducting independent evaluations of SCD studies.

How the WWC Deals with the Generality Criteria Described by Maggin et al. (2013)

Table 1 summarizes the generalization details captured by the WWC SCD *Review Guide*, which is publically available.[2] Trained and certified reviewers complete the

[2] The WWC SRG *Review Guide* is subject to change. A copy of the current *Review Guide* is available here: http://ies.ed.gov/ncee/wwc/DownloadSRG.aspx. The study *Review Guide* used by and/or referenced herein was developed by the U.S. Department of Education, Institute of Education Sciences through its What Works Clearinghouse project and was used by the authors with permission from the Institute of Education Sciences. Neither the Institute of Education Sciences nor its contractor administrators of the What Works Clearinghouse endorse the content herein.

Table 1 Summary of WWC SCD review guide items that capture generalization details

Maggin et al. (2013) generalization criterion	WWC review guide items (summarized) designed to capture related details
Baseline description	Do the data in the first baseline phase…document that (a) the concern is demonstrated and (b) …a clearly defined baseline pattern of responding…?[a]
	…describe the baseline condition as implemented in the study (including number of days/weeks/months, number of sessions, time per session[c]
Dependent variable operational definition	Does the study address at least one outcome in a domain relevant for the review protocol?[b]
	…describe all eligible outcomes reported and how they were measured[c]
	Are there outcomes that do not meet review requirements? If yes, provide a domain and brief description of why[c]
Independent variable description	…describe intervention condition as implemented in the study (including number of days/weeks/months, number of sessions, time per session)[c]
	…describe support needed for implementation[c]
	…describe maintenance phases if any (describe intervention and data patterns)[c]
Participant descriptions	Does the study meet the requirements for sample characteristics specified in the review protocol?[b]
	Does the study examine students in the age or grade range specified in the protocol?[b]
Setting descriptions	Does the study examine sample members in a location specified in the review protocol?[b]

[a] Review guide item used for visual analyses

[b] Review guide item used for study screening

[c] Review guide item used for report descriptions

Review Guide for each SCD study included in a specific review. The review consists of several phases. As the table shows, the information that allows one to make informed decisions regarding generalization is sometimes a matter of *screening*. The first phase of a review consists of preliminary screening as related to a specific protocol. Part of generalization entails understanding the goals of a given review, and thus, it is necessary to screen studies to determine whether an eligible intervention was examined; furthermore, it is necessary to assess whether dependent variables, settings, and participant characteristics meet protocol parameters. For example, a review protocol may specify that a treatment must be delivered in a K-12 school setting, and it is acceptable if treatment was offered in self-contained classrooms. However, the same treatment may not be of interest if applied in residential programs staffed by highly specialized personnel.

The second and the third phases of a review consist of evaluating the internal validity of a study. Briefly, the number of phase contrasts, number of data points per phase, evidence that outcomes were reliably measured, and whether the researchers actively manipulated the independent variable are all evaluated. Details may be further understood during visual analyses (e.g., understanding baseline performance relative to intervention phases). Again, we do not describe these two phases in detail (see Kratochwill et al. 2010, 2013) because the majority of information examined at this stage relates to the internal validity of a study rather than the generalization of findings or external validity. We briefly presented these phases to provide a more clear and logical sequence of a study *Review Guide*.

The fourth phase of a review consists of developing descriptions of study details. Study details consist of setting, design, participants' characteristics and sample size, operational definition of dependent variables and recording method, baseline and treatment conditions (e.g., number of sessions, length of intervention), the implementation agent and the training received prior to treatment, and treatment fidelity. As Table 1 shows, these study details address the generalization criteria discussed by Maggin et al. (2013).

An example illustrating the application of a review protocol to evaluate the empirical evidence from multiple studies, addressing both the internal validity and the external validity, is the recently released intervention report on *Repeated Reading* (WWC 2014), which incidentally was published after the Maggin et al. (2013) review. Although SCD evidence is a not a central concern in the *Repeated Reading* report, it does include a SCD that met the *Standards* as noted in Appendix E of the publication. The appendix contains information pertaining to generalizability criteria including baseline description, operational definitions of the dependent variable, participant descriptions, and setting.

The 5-3-20 Rule

Generalization details are not evaluated but rather described in detail so that consumers can make their own determinations about generalization. The WWC does, however, apply a threshold (i.e., the 5-3-20 rule) proposed by the *Standards* Panel. The *Standards* provide detailed information about this threshold, but as a summary, the current plan is to collate SCD studies into a single summary rating when (a) there are at least five SCD studies that meet WWC design standards (or standards with reservations), (b) the studies were conducted by at least three distinct research teams (housed in three different locations), and (c) the combined number of participants in the experiments is at least 20. For this last criterion, this may mean 20 individuals if there were, for example, 20 ABAB designs with only one student. At the same time, the ABAB designs may use aggregated units like classrooms. Furthermore, a multiple-baseline design may include several people, or sometimes one unit may be exposed to several baseline conditions (e.g., What is a student's behavior in Math, Reading and Science classes?).

These criteria are in fact somewhat arbitrary[3] but they are based on both expert judgment and logic and they are meant to be transparent. In addition, the threshold was vetted with SCD experts who did not serve on the Panel. The logic is simply that, if an effect has been replicated across 20 participants, by different teams, and different studies, then it is sufficiently robust to describe when addressing review questions about a set of treatments designed to meet the needs of a group of

[3] So are other common criteria such as setting p values at .05, see Cohen (1994).

participants (i.e., when the threshold is met the WWC will produce intervention reports solely on the basis of SCD evidence, even in absence of group studies that assess the impacts of the treatment being reviewed). The 5-3-20 threshold cannot of course yield guidance about where and how findings might generalize (indeed, 5-3-20 could conceivably be met by focusing on a very rare set of students exposed to treatments in unusual settings). Nevertheless, the approach does at least yield a marker for whether there is strong empirical evidence describing the effectiveness of a treatment. Because the 5-3-20 threshold is both new and somewhat arbitrary, it is our hope that it may be revisited after hard won review experience. It may well be overly stringent, or it could functionally suppress information that policy makers and practitioners hope the WWC might provide. Although this seems unlikely, it may prove to be too liberal a set of criteria.

Conclusion

Developing a widely accepted system for reviewing generalization procedures applicable to SCD studies is a difficult and complex process. It is possible to assess the evidence for whether treatment implementation is responsible for the observed change in a subject's performance because there is a well-defined common set of alternative explanations (i.e., threats) that SCDs can rule out. But without knowing the intended generalization point for evidence, it is difficult to make judgments along external validity lines. This circumstance is true of most studies designed to assess treatment effectiveness, including randomized controlled trials (see Hedges 2013). However, in the case of SCDs, evaluating external validity may be more because of a general (and erroneous) belief that SCD evidence cannot generalize when in fact, it can (Barlow et al. 2009). Therefore, there is much work to be done in terms of clarifying how SCD information can inform contexts outside the setting of the original study (Hitchcock et al. 2014). Indeed, this is a topic we hope to address in a separate article in which issues, such as replication, are featured. The key point of the current work is to clarify that the WWC review procedures do in fact gather information on all of the generalization criteria Maggin et al. (2013) list, but do not develop specific thresholds for assessing whether a criterion like "setting description" is good enough. The only threshold that has been developed is the 5-3-20 rule, and this may be revisited over time.

It will of course be easier for the field to understand the process after more reports are released by the WWC that illustrate how a review is conducted and how different generalization criteria are used to make inferences about generalization of findings based on multiple SCD studies examining a specific treatment. In the meantime, it is important to clarify how the system works so that the SCD research community understands the *Standards*. This understanding should facilitate use of future WWC reports and may influence both the conduct and the reporting of SCD studies. We hope that, in time, additional review work by reviewers, such as the WWC, will demonstrate the importance of SCD work to policymakers, practitioners, and researchers who will be able to test and refine ways of judging and communicating about matters of generalization. In the meantime, we take it as a good sign that there is more agreement than not about what sort of SCD details should be captured, assessed, and reported.

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